



CHE 109: Engineering Chemistry – I
Summer-2025

Basics of organic chemistry

Chapter 9

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Structures of Some Common Functional Groups

Name	Structure*	Name ending	Example
Alkene (double bond)		-ene	$\text{H}_2\text{C}=\text{CH}_2$ Ethene
Alkyne (triple bond)		-yne	$\text{HC}\equiv\text{CH}$ Ethyne
Arene (aromatic ring)		None	 Benzene
Halide	 (X = F, Cl, Br, I)	None	CH_3Cl Chloromethane
Alcohol		-ol	CH_3OH Methanol
Ether		ether	CH_3OCH_3 Dimethyl ether
Monophosphate		phosphate	$\text{CH}_3\text{OPO}_3^{2-}$ Methyl phosphate
Amine		-amine	CH_3NH_2 Methylamine
Imine (Schiff base)		None	$\text{CH}_3\text{C}(\text{NH})\text{CH}_3$ Acetone imine
Nitrile		-nitrile	$\text{CH}_3\text{C}\equiv\text{N}$ Ethanenitrile
Nitro		None	CH_3NO_2 Nitromethane
Thiol		-thiol	CH_3SH Methanethiol

*The bonds whose connections aren't specified are assumed to be attached to carbon or hydrogen atoms in the rest of the molecule.

Name	Structure*	Name ending	Example
Sulfide		sulfide	CH_3SCH_3 Dimethyl sulfide
Disulfide		disulfide	CH_3SSCH_3 Dimethyl disulfide
Carbonyl			
Aldehyde		-al	CH_3CHO Ethanal
Ketone		-one	CH_3COCH_3 Propanone
Carboxylic acid		-oic acid	CH_3COOH Ethanoic acid
Ester		-oate	$\text{CH}_3\text{COOCH}_3$ Methyl ethanoate
Amide		-amide	CH_3CONH_2 Ethanamide
Carboxylic acid anhydride		-oic anhydride	$\text{CH}_3\text{COOCOCH}_3$ Ethanoic anhydride
Carboxylic acid chloride		-oyl chloride	CH_3COCl Ethanoyl chloride

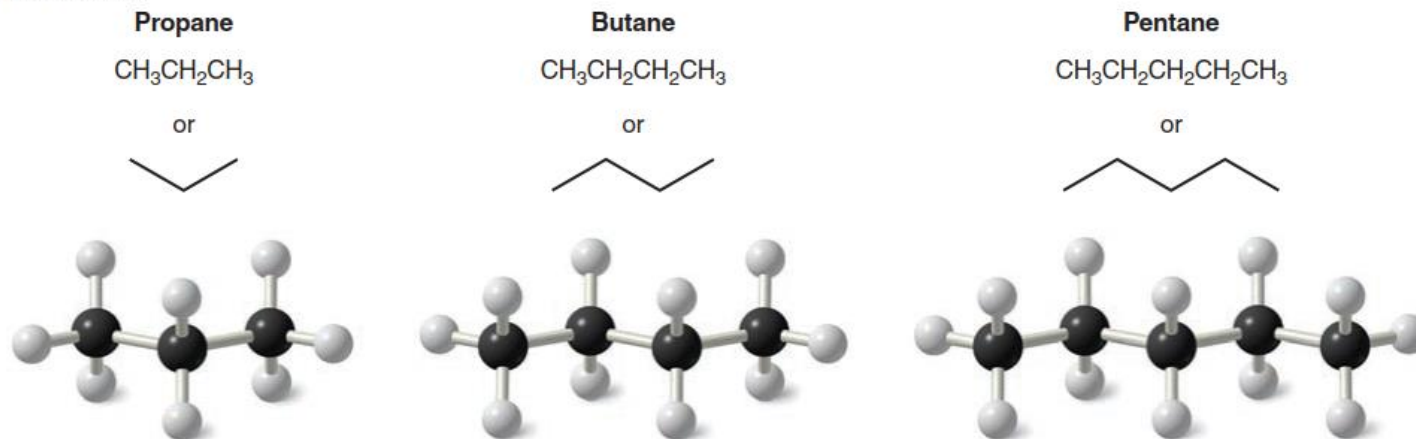
*The bonds whose connections aren't specified are assumed to be attached to carbon or hydrogen atoms in the rest of the molecule.

4.2 SHAPES OF ALKANES

A general tetrahedral orientation of groups—and thus sp^3 hybridization—is the rule for the carbon atoms of all alkanes and cycloalkanes. We can represent the shapes of **alkanes** as shown in Fig. 4.1.

Butane and pentane are examples of alkanes that are sometimes called “straight-chain” alkanes. One glance at three-dimensional models, however, shows that because of their tetrahedral carbon atoms the chains are zigzagged and not at all straight. Indeed,

FIGURE 4.1 Ball-and-stick models for three simple alkanes.



the structures that we have depicted in Fig. 4.1 are the straightest possible arrangements of the chains because rotations about the carbon–carbon single bonds produce arrangements that are even less straight. A better description is **unbranched**. This means that each carbon atom within the chain is bonded to no more than two other carbon atoms and that unbranched alkanes contain only primary and secondary carbon atoms. Primary, secondary, and tertiary carbon atoms were defined in Section 2.5.

Isobutane, isopentane, and neopentane (Fig. 4.2) are examples of branched-chain alkanes. In neopentane the central carbon atom is bonded to four carbon atoms.

Helpful Hint

You should build your own molecular models of the compounds in Figs. 4.1 and 4.2. View them from different perspectives and experiment with how their shapes change when you twist various bonds. Make drawings of your structures.

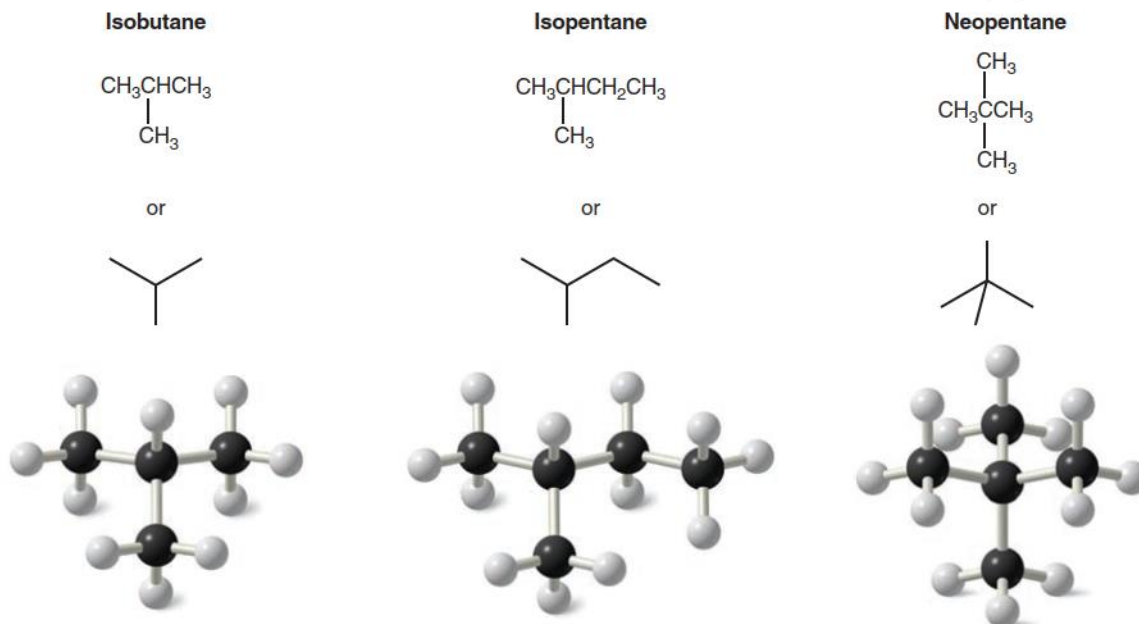
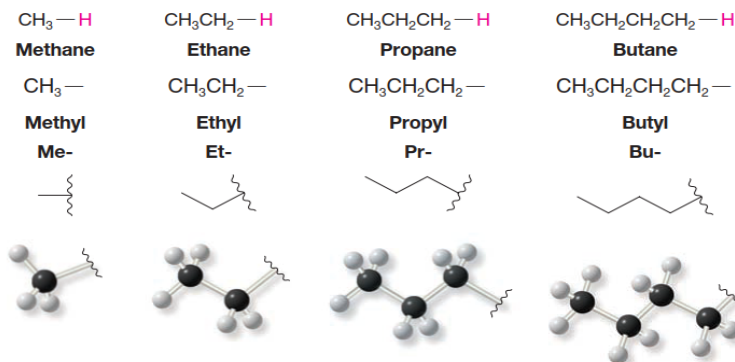


FIGURE 4.2 Ball-and-stick models for three branched-chain alkanes. In each of the compounds one carbon atom is attached to more than two other carbon atoms.

Butane and isobutane have the same molecular formula: C_4H_{10} . The two compounds have their atoms connected in a different order and are, therefore, **constitutional isomers** (Section 1.3). Pentane, isopentane, and neopentane are also constitutional isomers. They, too, have the same molecular formula (C_5H_{12}) but have different structures.

4.3A HOW TO Name Unbranched Alkyl Groups

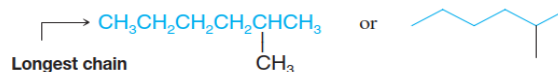
If we remove one hydrogen atom from an alkane, we obtain what is called an **alkyl group**. These alkyl groups have names that end in **-yl**. When the alkane is **unbranched**, and the hydrogen atom that is removed is a **terminal** hydrogen atom, the names are straightforward:



4.3B HOW TO Name Branched-Chain Alkanes

Branched-chain alkanes are named according to the following rules:

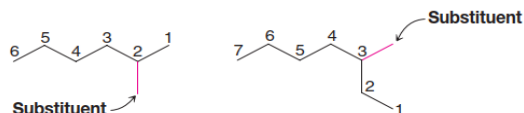
1. **Locate the longest continuous chain of carbon atoms; this chain determines the parent name for the alkane.** We designate the following compound, for example, as a *hexane* because the longest continuous chain contains six carbon atoms:



The longest continuous chain may not always be obvious from the way the formula is written. Notice, for example, that the following alkane is designated as a *heptane* because the longest chain contains seven carbon atoms:

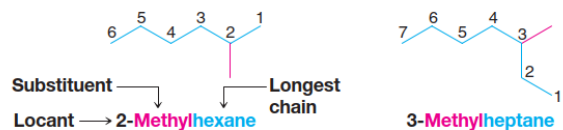


2. **Number the longest chain beginning with the end of the chain nearer the substituent.** Applying this rule, we number the two alkanes that we illustrated previously in the following way:

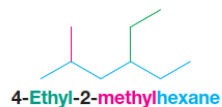


3. **Use the numbers obtained by application of rule 2 to designate the location of the substituent group.** The parent name is placed last, and the substituent group, preceded by the number designating its location on the chain, is placed first. Numbers

are separated from words by a hyphen. Our two examples are 2-methylhexane and 3-methylheptane, respectively:

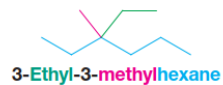


4. When two or more substituents are present, give each substituent a number corresponding to its location on the longest chain. For example, we designate the following compound as 4-ethyl-2-methylhexane:



The substituent groups should be listed *alphabetically* (i.e., ethyl before methyl).^{*} In deciding on alphabetical order, disregard multiplying prefixes such as “di” and “tri.”

5. When two substituents are present on the same carbon atom, use that number twice:

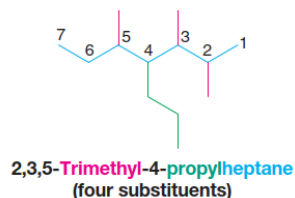


6. When two or more substituents are identical, indicate this by the use of the prefixes **di-**, **tri-**, **tetra-**, and so on. Then make certain that each and every substituent has a number. Commas are used to separate numbers from each other:

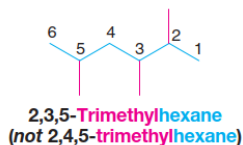


Application of these six rules allows us to name most of the alkanes that we shall encounter. Two other rules, however, may be required occasionally:

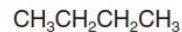
7. When two chains of equal length compete for selection as the parent chain, choose the chain with the greater number of substituents:



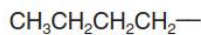
8. When branching first occurs at an equal distance from either end of the longest chain, choose the name that gives the lower number at the first point of difference:



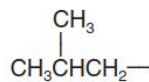
Four-Carbon Groups



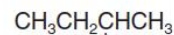
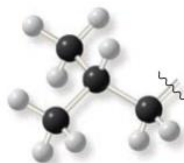
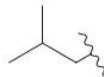
Butane



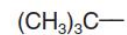
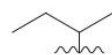
Butyl



Isobutyl



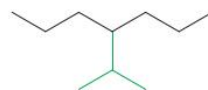
sec-Butyl



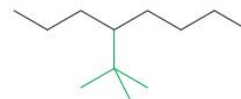
tert-Butyl (or *t*-Bu)



The following examples show how the names of these groups are employed:



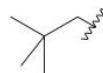
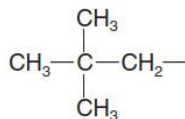
4-(1-Methylethyl)heptane or 4-isopropylheptane



4-(1,1-Dimethylethyl)octane or 4-tert-butyloctane

The common names **isopropyl**, **isobutyl**, **sec-butyl**, and **tert-butyl** are approved by the IUPAC for the unsubstituted groups, and they are still very frequently used. You should learn these groups so well that you can recognize them any way they are written. In deciding on alphabetical order for these groups you should disregard structure-defining prefixes that are written in italics and separated from the name by a hyphen. Thus *tert*-butyl precedes ethyl, but ethyl precedes isobutyl.*

There is one five-carbon group with an IUPAC approved common name that you should also know: the 2,2-dimethylpropyl group, commonly called the **neopentyl group**.



2,2-Dimethylpropyl or neopentyl group

The simple branched alkyl groups are usually known by common names. The isopropyl and isobutyl groups have a characteristic “iso” $(\text{CH}_3)_2\text{CH}$ grouping, just as in isobutane.

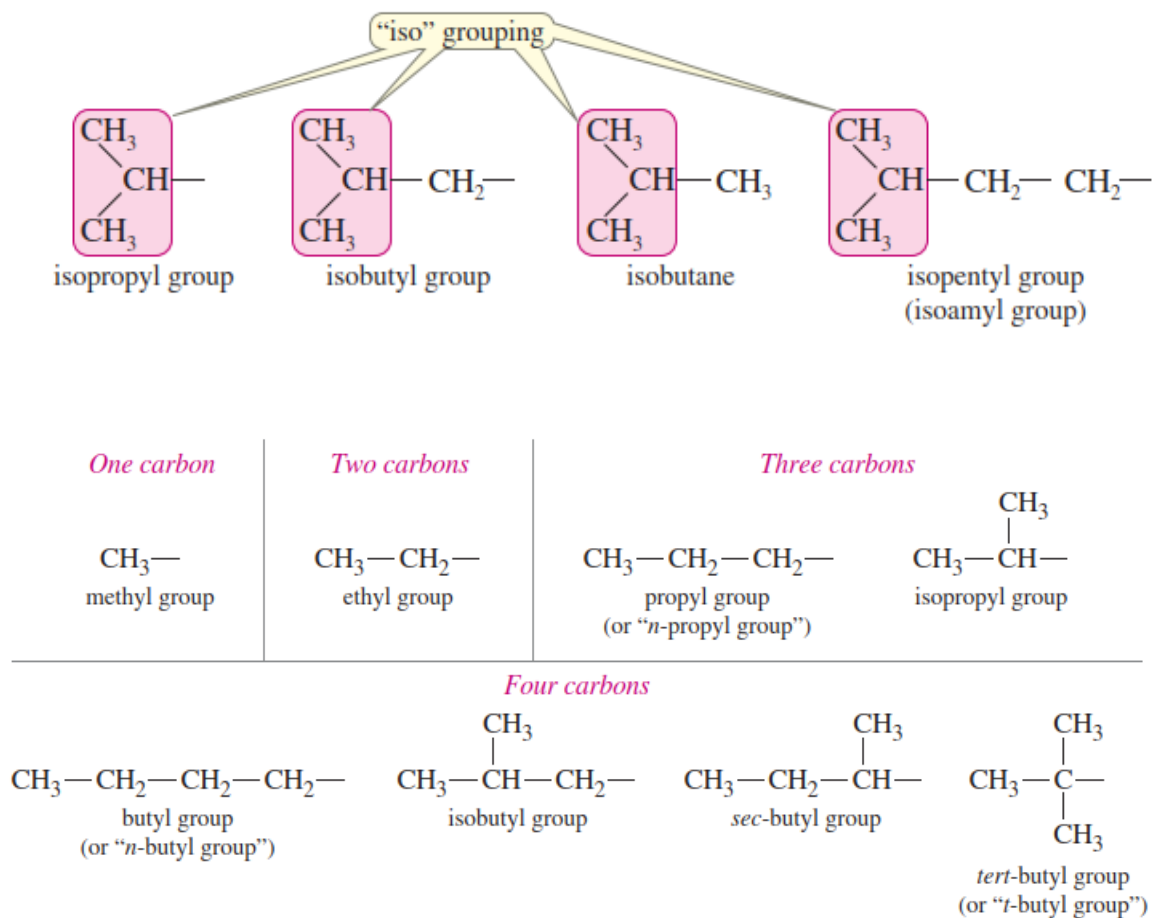
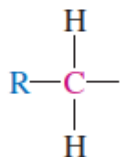
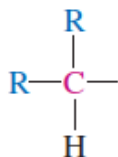


FIGURE 3-2 Some common alkyl groups.

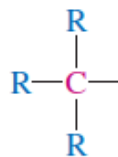
The names of the *secondary*-butyl (*sec*-butyl) and *tertiary*-butyl (*tert*-butyl or *t*-butyl) groups are based on the **degree of alkyl substitution** of the carbon atom attached to the main chain. In the *sec*-butyl group, the carbon atom bonded to the main chain is **secondary** (2°), or bonded to two other carbon atoms. In the *tert*-butyl group, it is **tertiary** (3°), or bonded to three other carbon atoms. In both the *n*-butyl group and the isobutyl group, the carbon atoms bonded to the main chain are **primary** (1°), bonded to only one other carbon atom.



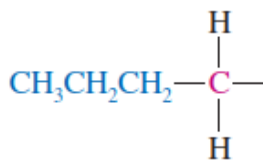
a *primary* (1°) carbon



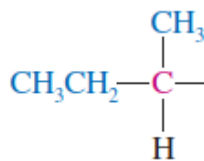
a *secondary* (2°) carbon



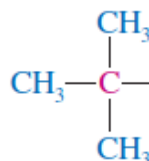
a *tertiary* (3°) carbon



n-butyl group (1°)



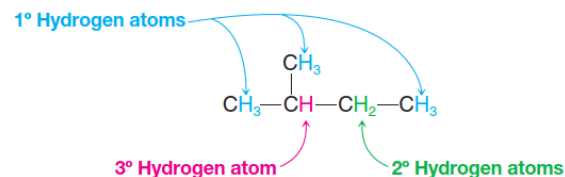
sec-butyl group (2°)



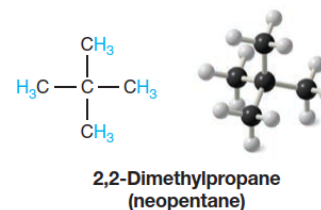
tert-butyl group (3°)

4.3D HOW TO Classify Hydrogen Atoms

The hydrogen atoms of an alkane are classified on the basis of the carbon atom to which they are attached. A hydrogen atom attached to a primary carbon atom is a primary (1°) hydrogen atom, and so forth. The following compound, 2-methylbutane, has primary, secondary (2°), and tertiary (3°) hydrogen atoms:



On the other hand, 2,2-dimethylpropane, a compound that is often called **neopentane**, has only primary hydrogen atoms:



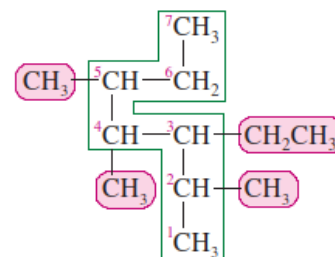
When two or more substituents are present, list them in alphabetical order. When two or more of the *same* alkyl substituent are present, use the prefixes *di-*, *tri-*, *tetra-*, etc., to avoid having to name the alkyl group twice. Include a position number for each substituent, even if it means repeating a number more than once.

di- means 2
tri- means 3
tetra- means 4

penta- means 5
hexa- means 6
hepta- means 7

octa- means 8
nona- means 9
deca- means 10

Using this rule, we can construct names for some complicated structures. Let's finish naming the heptane shown again here in the margin. This compound has an ethyl group on C3 and three methyl groups on C2, C4, and C5. List the ethyl group alphabetically before the methyl groups, and give each of the four substituents a location number.



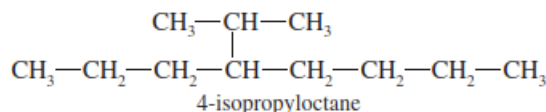
3-ethyl-2,4,5-trimethylheptane

SOLVED PROBLEM 3-1

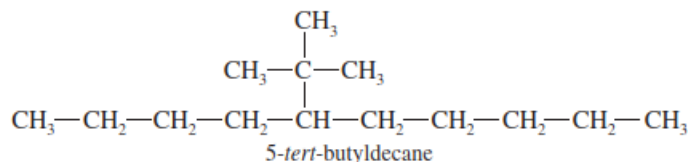
Give the structures of 4-isopropyloctane and 5-*tert*-butyldecane.

SOLUTION

The base name of 4-isopropyloctane is octane, so it has a main chain of eight carbons. There must be an isopropyl group on the fourth carbon from the end.

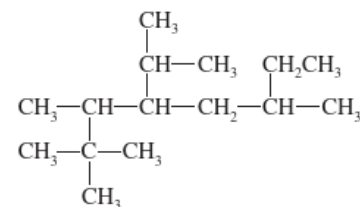


The base name of 5-*tert*-butyldecane is decane, so it has a main chain of ten carbons. There must be a *tert*-butyl group on the fifth carbon from the end.



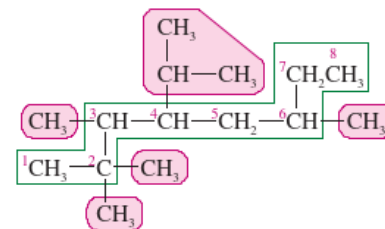
SOLVED PROBLEM 3-2

Give a systematic (IUPAC) name for the following compound.



SOLUTION

The longest carbon chain contains eight carbon atoms, so this compound is named as an octane. Numbering from left to right gives the first branch on C2; numbering from right to left gives the first branch on C3, so we number from left to right.

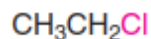


There are four methyl groups: two on C2, one on C3, and one on C6. These four groups will be listed as "2,2,3,6-tetramethyl . . ." There is an isopropyl group on C4. Listing the isopropyl group and the methyl groups alphabetically, we have

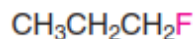
4-isopropyl-2,2,3,6-tetramethyloctane

4.3E HOW TO Name Alkyl Halides

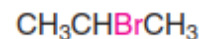
Alkanes bearing halogen substituents are named in the IUPAC substitutive system as haloalkanes:



Chloroethane

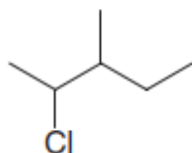


1-Fluoropropane

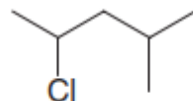


2-Bromopropane

- When the parent chain has both a halo and an alkyl substituent attached to it, number the chain from the end nearer the first substituent, regardless of whether it is halo or alkyl. If two substituents are at equal distance from the end of the chain, then number the chain from the end nearer the substituent that has alphabetical precedence:

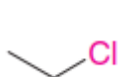


2-Chloro-3-methylpentane



2-Chloro-4-methylpentane

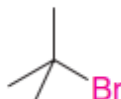
Common names for many simple haloalkanes are still widely used, however. In this common nomenclature system, called **functional class nomenclature**, haloalkanes are named as alkyl halides. (The following names are also accepted by the IUPAC.)



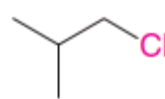
**Ethyl
chloride**



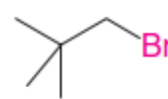
**Isopropyl
bromide**



**tert-Butyl
bromide**



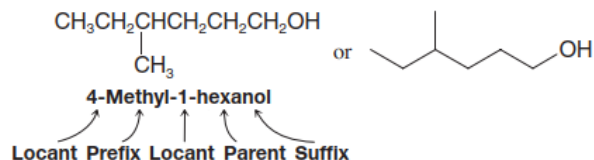
**Isobutyl
chloride**



**Neopentyl
bromide**

4.3F HOW TO Name Alcohols

In what is called IUPAC **substitutive nomenclature**, a name may have as many as four features: **locants**, **prefixes**, **parent compound**, and **suffixes**. Consider the following compound as an illustration without, for the moment, being concerned as to how the name arises:



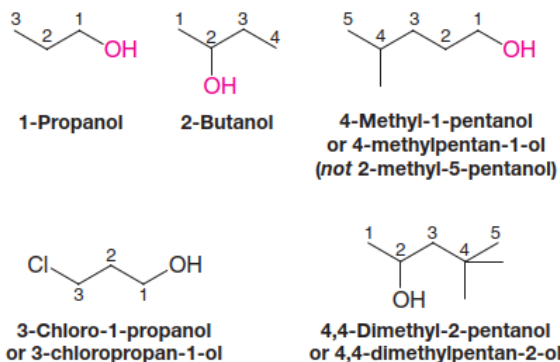
The *locant 4-* tells that the substituent **methyl** group, named as a *prefix*, is attached to the *parent compound* at C4. The parent compound contains six carbon atoms and no multiple bonds, hence the parent name **hexane**, and it is an alcohol; therefore it has the *suffix -ol*. The *locant 1-* tells that C1 bears the hydroxyl group. **In general, numbering of the chain always begins at the end nearer the group named as a suffix.**

The locant for a suffix (whether it is for an alcohol or another functional group) may be placed before the parent name as in the above example or, according to a 1993 IUPAC revision of the rules, immediately before the suffix. Both methods are IUPAC approved. Therefore, the above compound could also be named **4-methylhexan-1-ol**.

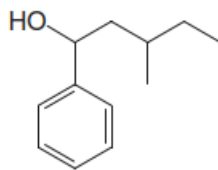
The following procedure should be followed in giving alcohols IUPAC substitutive names:

1. Select the longest continuous carbon chain *to which the hydroxyl is directly attached*. Change the name of the alkane corresponding to this chain by dropping the final -e and adding the suffix -ol.
2. Number the longest continuous carbon chain so as to give the carbon atom bearing the hydroxyl group the lower number. Indicate the position of the hydroxyl group by using this number as a locant; indicate the positions of other substituents (as prefixes) by using the numbers corresponding to their positions along the carbon chain as locants.

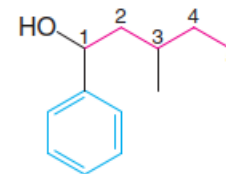
The following examples show how these rules are applied:



Give an IUPAC name for the compound shown.



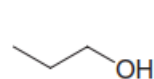
STRATEGY AND ANSWER: We find that the longest carbon chain (in red at right) has five carbons and it bears a hydroxyl group on the first carbon. So we name this part of the molecule as a 1-pentanol. There is a phenyl group on carbon-1 and a methyl group on carbon-3, so the full name is 3-methyl-1-phenyl-1-pentanol.



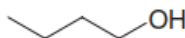
Draw bond-line formulas and give IUPAC substitutive names for all of the isomeric alcohols with the formulas **(a)** $C_4H_{10}O$ and **(b)** $C_5H_{12}O$.

PRACTICE PROBLEM 4.7

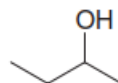
Simple alcohols are often called by *common* functional class names that are also approved by the IUPAC. We have seen several examples already (Section 2.6). In addition to *methyl alcohol*, *ethyl alcohol*, and *isopropyl alcohol*, there are several others, including the following:



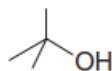
Propyl alcohol



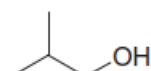
Butyl alcohol



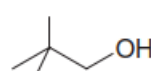
sec-Butyl alcohol



tert-Butyl alcohol

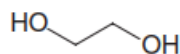


Isobutyl alcohol



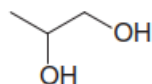
Neopentyl alcohol

Alcohols containing two hydroxyl groups are commonly called **glycols**. In the IUPAC substitutive system they are named as **diols**:



Substitutive
Common

1,2-Ethanediol
or ethane-1,2-diol
Ethylene glycol



1,2-Propanediol
or propane-1,2-diol
Propylene glycol

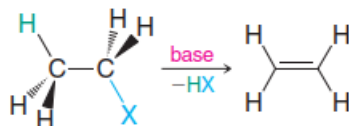


1,3-Propanediol
or propane-1,3-diol
Trimethylene glycol

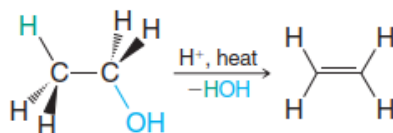
7.5 SYNTHESIS OF ALKENES VIA ELIMINATION REACTIONS

Elimination reactions are the most important means for synthesizing alkenes. In this chapter we shall study two methods for alkene synthesis based on elimination reactions: dehydrohalogenation of alkyl halides and dehydration of alcohols.

Dehydrohalogenation of Alkyl Halides (Sections 6.15, 6.16, and 7.6)



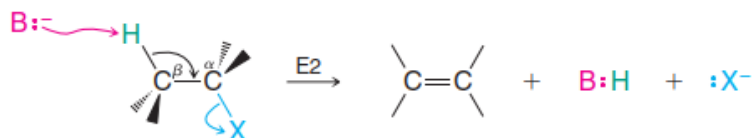
Dehydration of Alcohols (Sections 7.7 and 7.8)



7.6 DEHYDROHALOGENATION OF ALKYL HALIDES

- The best reaction conditions to use when synthesizing an alkene by **dehydrohalogenation** are those that promote an E2 mechanism.

In an E2 mechanism, a base removes a β hydrogen from the β carbon, as the double bond forms and a leaving group departs from the α carbon.



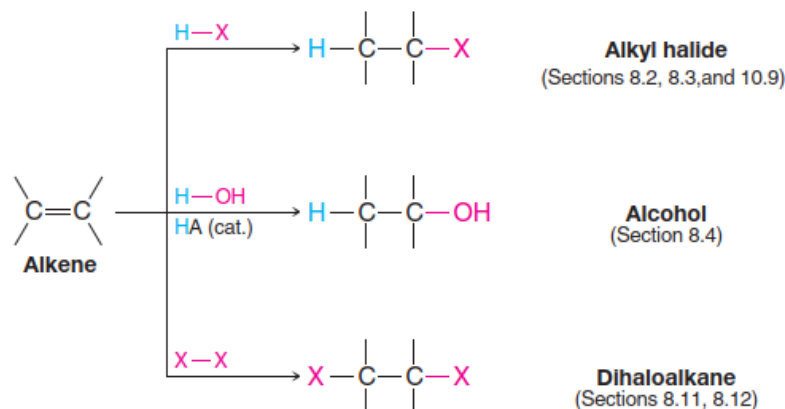
Reaction conditions that favor elimination by an E1 mechanism should be avoided because the results can be too variable. The carbocation intermediate that accompanies an E1 reaction can undergo rearrangement of the carbon skeleton, as we shall see in Section 7.8, and it can also undergo substitution by an S_N1 mechanism, which competes strongly with formation of products by an E1 path.

8.1 ADDITION REACTIONS OF ALKENES

We have already studied one addition reaction of alkenes—hydrogenation—in which a hydrogen atom is added at each end of a double (or triple) bond. In this chapter we shall study other alkene addition reactions that do not involve the same mechanism as hydrogenation. We can depict this type of reaction generally, using E for an electrophilic portion of a reagent and Nu for a nucleophilic portion, as follows.



Some specific reactions of this type that we shall study in this chapter include addition of hydrogen halides, sulfuric acid, water (in the presence of an acid catalyst), and halogens. Later we shall also study some specialized reagents that undergo addition reactions with alkenes.

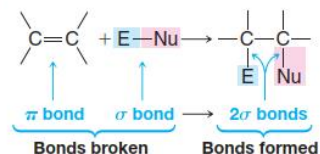


8.1A HOW TO Understand Additions to Alkenes

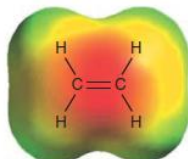
Two characteristics of the double bond help us understand why these addition reactions occur:

1. An addition reaction results in the conversion of one π bond and one σ bond (Sections 1.12 and 1.13) into two σ bonds. The result of this change is usually energetically favorable. The energy released in making two σ bonds exceeds that needed to break

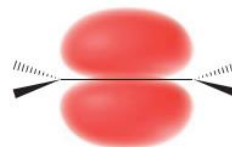
one σ bond and one π bond (because π bonds are weaker), and, therefore, addition reactions are usually exothermic:



- The electrons of the π bond are exposed. Because the π bond results from overlapping p orbitals, the π electrons lie above and below the plane of the double bond:



An electrostatic potential map for ethene shows the higher density of negative charge in the region of the π bond.



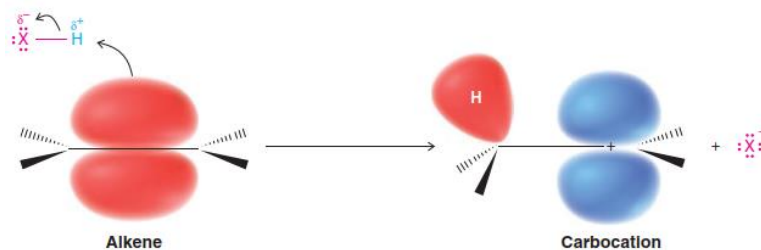
The electron pair of the π bond is distributed throughout both lobes of the π molecular orbital.

Electrophilic Addition

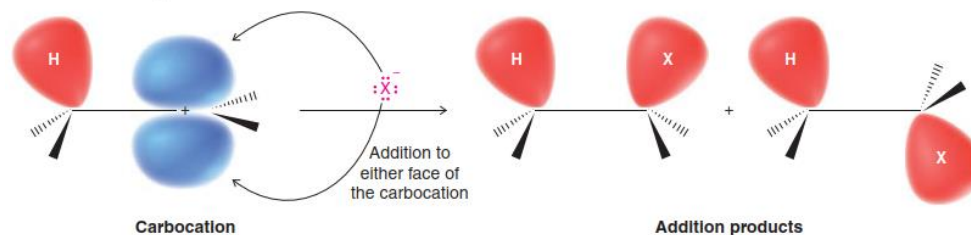
- Electrons in the π bond of alkenes react with electrophiles.
- Electrophiles** are electron-seeking reagents. They have the property of being **electrophilic**.

Electrophiles include proton donors such as Brønsted-Lowry acids, neutral reagents such as bromine (because it can be polarized so that one end is positive), and Lewis acids such as BH_3 , BF_3 , and AlCl_3 . Metal ions that contain vacant orbitals—the silver ion (Ag^+), the mercuric ion (Hg^{2+}), and the platinum ion (Pt^{2+}), for example—also act as electrophiles.

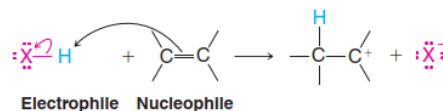
Hydrogen halides, for example, react with alkenes by accepting a pair of electrons from the π bond to form a σ bond between the hydrogen and one of the carbon atoms, with loss of the halide ion. This leaves a vacant p orbital and a + charge on the other carbon. The overall result is the formation of a carbocation and a halide ion from the alkene and HX :



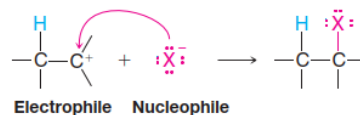
Being highly reactive, the carbocation may then combine with the halide ion by accepting one of its electron pairs:



Electrophiles Are Lewis Acids Electrophiles are molecules or ions that can accept an electron pair. Nucleophiles are molecules or ions that can furnish an electron pair (i.e., Lewis bases). Any reaction of an electrophile also involves a nucleophile. In the protonation of an alkene the electrophile is the proton donated by an acid; the nucleophile is the alkene:

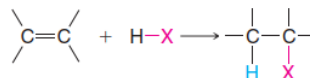


In the next step, the reaction of the carbocation with a halide ion, the carbocation is the electrophile and the halide ion is the nucleophile:



8.2 ELECTROPHILIC ADDITION OF HYDROGEN HALIDES TO ALKENES: MECHANISM AND MARKOVNIKOV'S RULE

Hydrogen halides (HI, HBr, HCl, and HF) add to the double bond of alkenes:



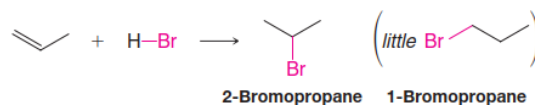
These additions are sometimes carried out by dissolving the hydrogen halide in a solvent, such as acetic acid or CH_2Cl_2 , or by bubbling the gaseous hydrogen halide directly into the alkene and using the alkene itself as the solvent. HF is prepared as polyhydrogen fluoride in pyridine.

- The order of reactivity of the hydrogen halides in alkene addition is

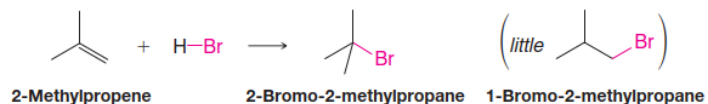


Unless the alkene is highly substituted, HCl reacts so slowly that the reaction is not one that is useful as a preparative method. HBr adds readily, but as we shall learn in Section 10.9, unless precautions are taken, the reaction may follow an alternate course.

The addition of HX to an unsymmetrical alkene could conceivably occur in two ways. In practice, however, one product usually predominates. The addition of HBr to propene, for example, could conceivably lead to either 1-bromopropane or 2-bromopropane. The main product, however, is 2-bromopropane:



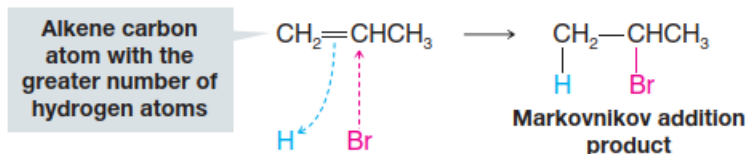
When 2-methylpropene reacts with HBr, the main product is 2-bromo-2-methylpropane, not 1-bromo-2-methylpropane:



Consideration of many examples like this led the Russian chemist Vladimir Markovnikov in 1870 to formulate what is now known as Markovnikov's rule.

- One way to state **Markovnikov's rule** is to say that *in the addition of HX to an alkene, the hydrogen atom adds to the carbon atom of the double bond that already has the greater number of hydrogen atoms.**

The addition of HBr to propene is an illustration:

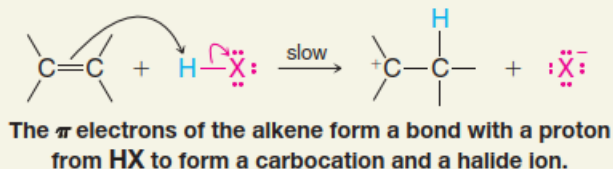


Reactions that illustrate Markovnikov's rule are said to be *Markovnikov additions*.

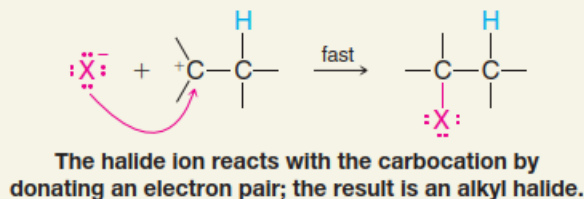
A mechanism for addition of a hydrogen halide to an alkene involves the following two steps:

[A MECHANISM FOR THE REACTION — Addition of a Hydrogen Halide to an Alkene]

Step 1



Step 2



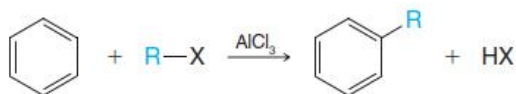
The important step—because it is the **rate-determining step**—is step 1. In step 1 the alkene donates a pair of electrons to the proton of the hydrogen halide and forms a carbocation. This step (Fig. 8.1) is highly endergonic and has a high free energy of activation. Consequently, it takes place slowly. In step 2 the highly reactive carbocation stabilizes itself by combining with a halide ion. This exergonic step has a very low free energy of activation and takes place very rapidly.

15.6 FRIEDEL–CRAFTS ALKYLATION

Charles Friedel, a French chemist, and his American collaborator, James M. Crafts, discovered new methods for the preparation of alkylbenzenes (ArR) and acylbenzenes (ArCOR) in 1877. These reactions are now called the Friedel–Crafts **alkylation** and

acylation reactions. We shall study the Friedel–Crafts alkylation reaction here and take up the **Friedel–Crafts acylation** reaction in Section 15.7.

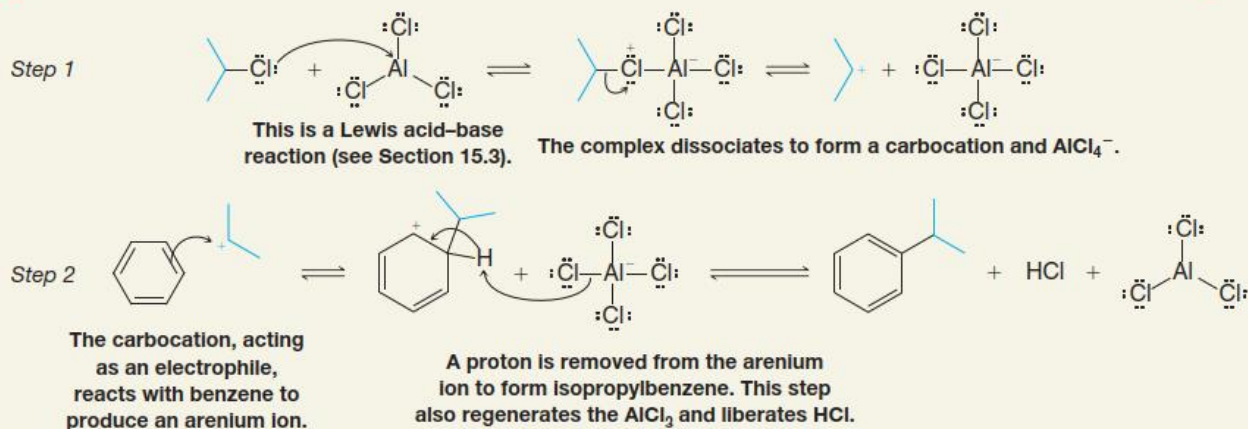
- The following is a general equation for a **Friedel–Crafts alkylation** reaction:



- The mechanism for the reaction starts with the formation of a carbocation.
- The carbocation then acts as an electrophile and attacks the benzene ring to form an arenium ion.
- The arenium ion then loses a proton.

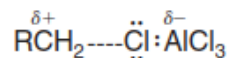
This mechanism is illustrated below using 2-chloropropane and benzene.

[A MECHANISM FOR THE REACTION — Friedel–Crafts Alkylation]



- When $R-X$ is a primary halide, a simple carbocation probably does not form. Instead, the aluminum chloride forms a complex with the alkyl halide, and this complex acts as the electrophile.

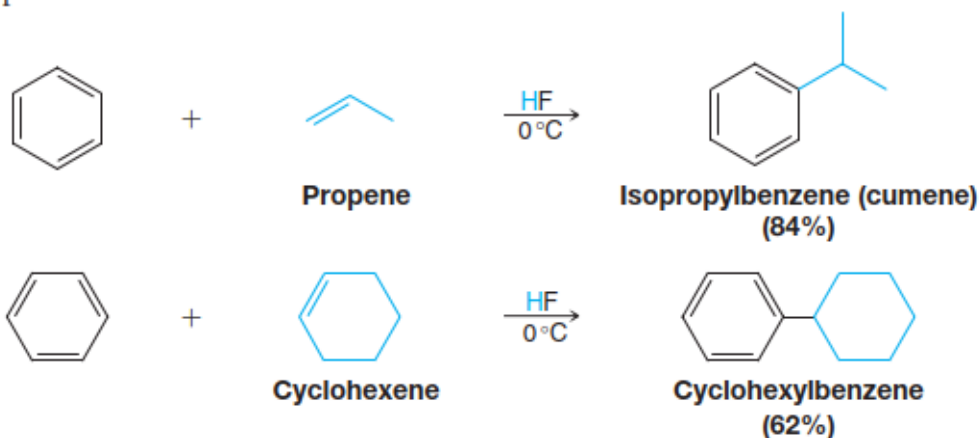
The complex is one in which the carbon–halogen bond is nearly broken—and one in which the carbon atom has a considerable positive charge:



Even though this complex is not a simple carbocation, it acts as if it were and it transfers a positive alkyl group to the aromatic ring.

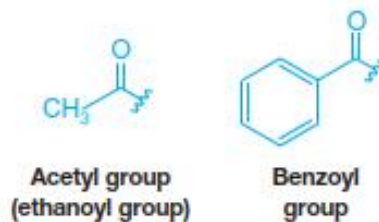
- These complexes react so much like carbocations that they also undergo typical carbocation rearrangements (Section 15.8).
- Friedel–Crafts alkylations are not restricted to the use of alkyl halides and aluminum chloride. Other pairs of reagents that form carbocations (or species like carbocations) may be used in Friedel–Crafts alkylations as well.

These possibilities include the use of a mixture of an alkene and an acid:

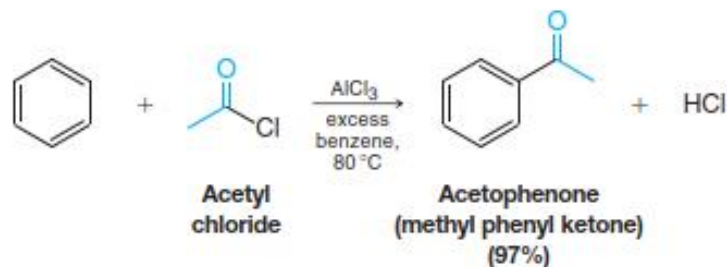


15.7 FRIEDEL-CRAFTS ACYLATION

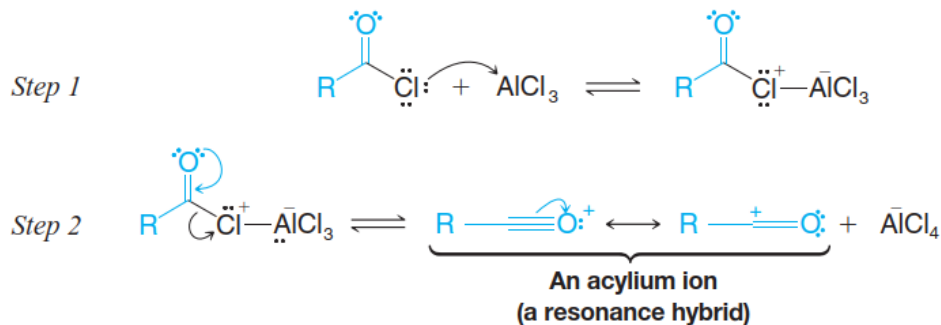
The $\text{R}-\text{C}(=\text{O})-$ group is called an **acyl group**, and a reaction whereby an acyl group is introduced into a compound is called an **acylation** reaction. Two common acyl groups are the acetyl group and the benzoyl group. (The benzoyl group should not be confused with the benzyl group, $-\text{CH}_2\text{C}_6\text{H}_5$; see Section 14.2.)



The **Friedel–Crafts acylation** reaction is often carried out by treating the aromatic compound with an **acyl halide** (often an acyl chloride). Unless the aromatic compound is one that is highly reactive, the reaction requires the addition of at least one equivalent of a Lewis acid (such as AlCl_3) as well. The product of the reaction is an aryl ketone:

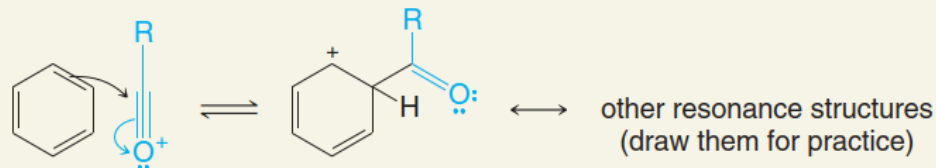


In most Friedel–Crafts acylations the electrophile appears to be an **acylium ion** formed from an acyl halide in the following way:



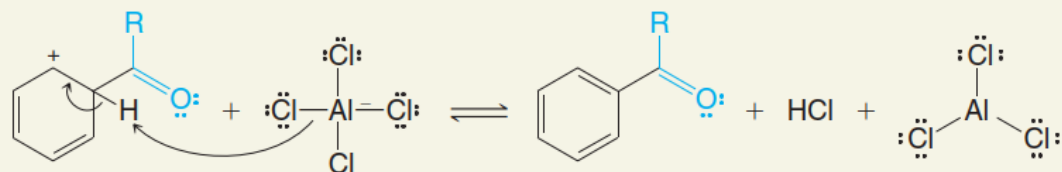
[A MECHANISM FOR THE REACTION – Friedel–Crafts Acylation]

Step 3



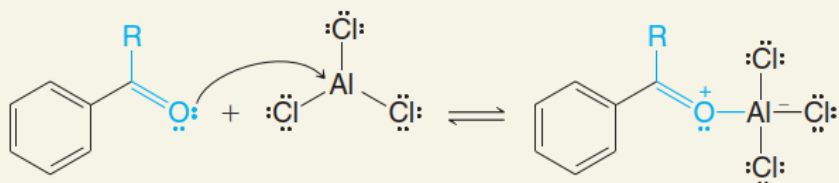
The acylium ion, acting as an electrophile,
reacts with benzene to form the arenium ion.

Step 4



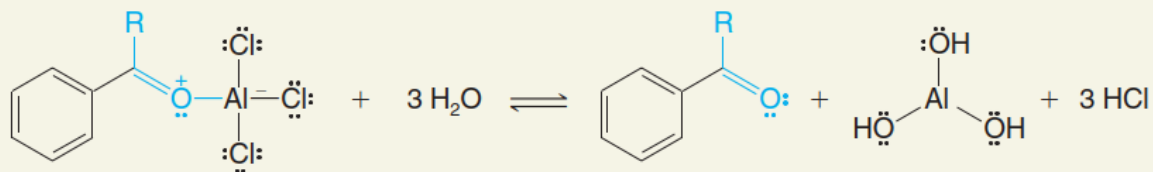
A proton is removed from the arenium ion, forming the aryl ketone.

Step 5



The ketone, acting as a Lewis base, reacts with
aluminum chloride (a Lewis acid) to form a complex.

Step 6



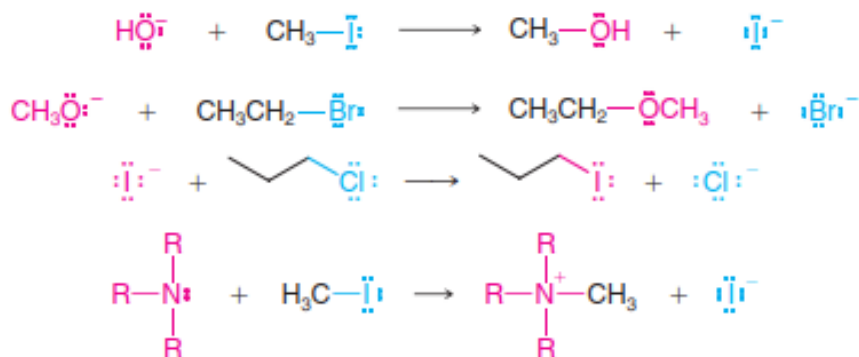
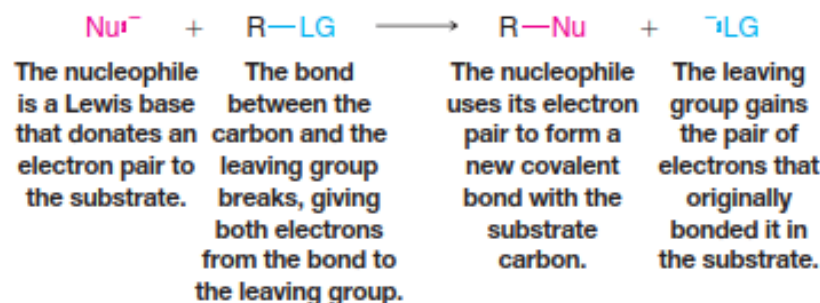
Treating the complex with water liberates the ketone and hydrolyzes the Lewis acid.

6.2 NUCLEOPHILIC SUBSTITUTION REACTIONS

Nucleophilic substitution reactions are among the most fundamental types of organic reactions. In a **nucleophilic substitution reaction** a nucleophile (**Nu:**) displaces a leaving group (**LG**) in the molecule that undergoes the substitution (the substrate).

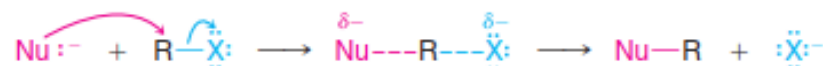
- The **nucleophile** is always a Lewis base, and it may be negatively charged or neutral.
- The **leaving group** is always a species that takes a pair of electrons with it when it departs.

Often the **substrate** is an alkyl halide ($\text{R}-\ddot{\text{X}}:$) and the leaving group is a halide anion ($:\ddot{\text{X}}:^-$). The following equations include a generic nucleophilic substitution reaction and some specific examples.

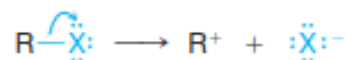


In nucleophilic substitution reactions the bond between the substrate carbon and the leaving group undergoes *heterolytic* bond cleavage. The unshared electron pair of the nucleophile forms the new bond to the carbon atom.

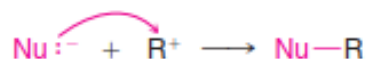
A key question we shall want to address later in this chapter is this: when does the bond between the leaving group and the carbon break? Does it break at the same time that the new bond between the nucleophile and carbon forms, as shown below?



Or, does the bond to the leaving group break first?



Followed by

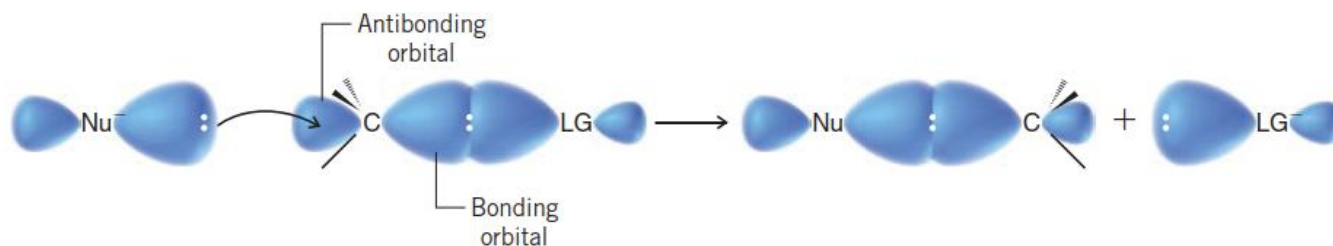


We shall find in Sections 6.9 and 6.14A that the answer depends greatly on the structure of the substrate.

6.6 A MECHANISM FOR THE S_N2 REACTION

A schematic representation of orbitals involved in an S_N2 reaction—based on ideas proposed by Edward D. Hughes and Sir Christopher Ingold in 1937—is outlined below.

*In general, the overall order of a reaction is equal to the sum of the exponents a and b in the rate equation $\text{Rate} = k[\text{A}]^a [\text{B}]^b$. If in some other reaction, for example, we found that $\text{Rate} = k[\text{A}]^2 [\text{B}]$, then we would say that the reaction is second order with respect to $[\text{A}]$, first order with respect to $[\text{B}]$, and third order overall.



According to this mechanism:

- The nucleophile approaches the carbon bearing the leaving group from the **back side**, that is, from the side directly opposite the leaving group.

The orbital that contains the electron pair of the nucleophile (its highest occupied molecular orbital, or HOMO) begins to overlap with an empty orbital (the lowest unoccupied molecular orbital, or LUMO) of the carbon atom bearing the leaving group. As the reaction progresses, the bond between the nucleophile and the carbon atom strengthens, and the bond between the carbon atom and the leaving group weakens.

- As the nucleophile forms a bond and the leaving group departs, the substrate carbon atom undergoes **inversion***—its tetrahedral bonding configuration is turned inside out.

The formation of the bond between the nucleophile and the carbon atom provides most of the energy necessary to break the bond between the carbon atom and the leaving group. We can represent this mechanism with chloromethane and hydroxide ion as shown in the “Mechanism for the S_N2 Reaction” box below.

- The S_N2 reaction proceeds in a single step (without any intermediates) through an unstable arrangement of atoms called the **transition state**.

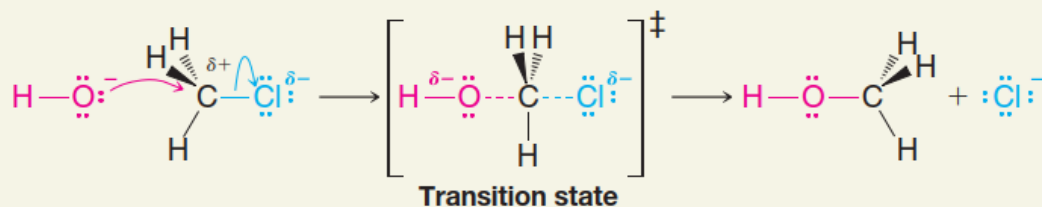
A MECHANISM FOR THE REACTION

Mechanism for the S_N2 Reaction

Reaction



Mechanism



The negative hydroxide ion brings a pair of electrons to the partially positive carbon from the back side with respect to the leaving group. The chlorine begins to move away with the pair of electrons that bonded it to the carbon.

In the transition state, a bond between oxygen and carbon is partially formed and the bond between carbon and chlorine is partially broken. The configuration of the carbon atom begins to invert.

Now the bond between the oxygen and carbon has formed and the chloride ion has departed. The configuration of the carbon has inverted.

*Considerable evidence had appeared in the years prior to Hughes and Ingold's 1937 publication indicating that in reactions like this an inversion of configuration of the carbon bearing the leaving group takes place. The first observation of such an inversion was made by the Latvian chemist Paul Walden in 1896, and such inversions are called **Walden inversions** in his honor. We shall study this aspect of the S_N2 reaction further in Section 6.8.

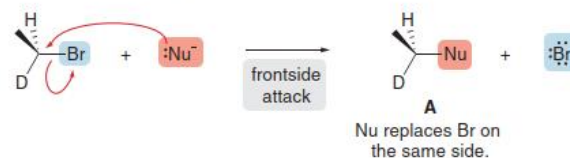
7.11C Stereochemistry of the S_N2 Reaction

From what direction does the nucleophile approach the substrate in an S_N2 reaction? There are two possibilities.

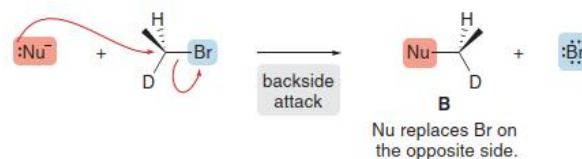
- **Frontside attack:** The nucleophile approaches from the *same* side as the leaving group.
- **Backside attack:** The nucleophile approaches from the side *opposite* the leaving group.

The results of frontside and backside attack of a nucleophile are illustrated with CH₃CH(D)Br as substrate and the general nucleophile :Nu⁻. This substrate has the leaving group bonded to a stereogenic center, thus allowing us to see the structural difference that results when the nucleophile attacks from two different directions.

In **frontside attack**, the nucleophile approaches from the **same** side as the leaving group, forming **A**. In this example, the leaving group was drawn on the right, so the nucleophile attacks from the right, and all other groups remain in their original positions. Because the nucleophile and leaving group are in the same position relative to the other three groups on carbon, frontside attack results in **retention of configuration** around the stereogenic center.



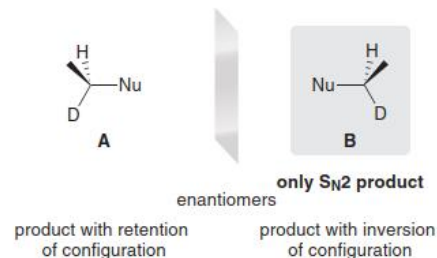
In **backside attack**, the nucleophile approaches from the **opposite** side to the leaving group, forming **B**. In this example, the leaving group was drawn on the right, so the nucleophile attacks from the left. Because the nucleophile and leaving group are in the opposite position relative to the other three groups on carbon, backside attack results in **inversion of configuration** around the stereogenic center.



Inversion of configuration in an S_N2 reaction is often called **Walden inversion**, after Latvian chemist Dr. Paul Walden, who first observed this process in 1896.

Backside attack occurs in all S_N2 reactions, but we can observe this change only when the leaving group is bonded to a stereogenic center.

The products of frontside and backside attack are *different* compounds. **A** and **B** are stereoisomers that are nonsuperimposable—they are **enantiomers**.



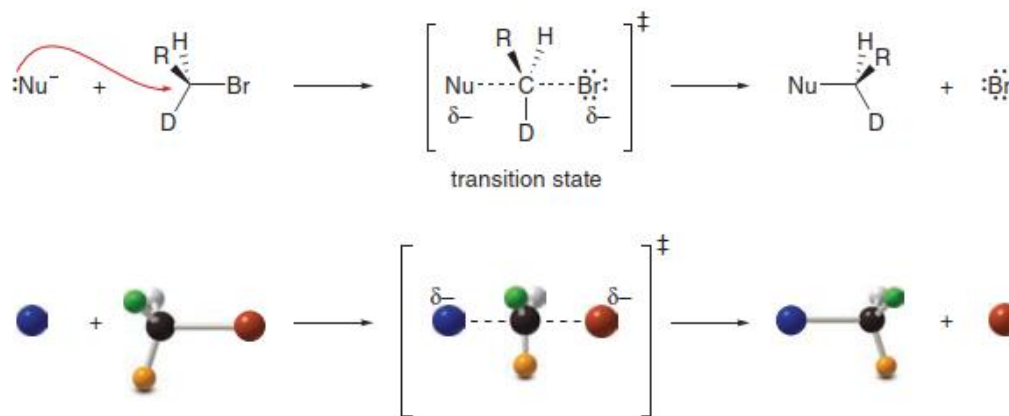
Which product is formed in an S_N2 reaction? When the stereochemistry of the product is determined, **only B, the product of backside attack, is formed.**

- All S_N2 reactions proceed with backside attack of the nucleophile, resulting in *inversion* of configuration at a stereogenic center.

One explanation for backside attack is based on an electronic argument. Both the nucleophile and leaving group are electron rich and these like charges repel each other. Backside attack

keeps these two groups as far away from each other as possible. In the transition state, the nucleophile and leaving group are 180° away from each other, and the other three groups around carbon occupy a plane, as illustrated in Figure 7.8.

Figure 7.8
Stereochemistry of
the S_N2 reaction

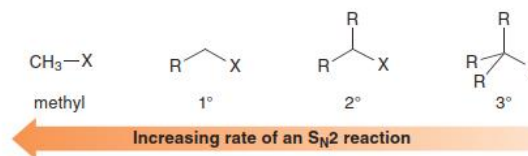


- :Nu^- and Br^- are 180° away from each other, on either side of a plane containing R, H, and D.

7.11D The Identity of the R Group

How does the rate of an S_N2 reaction change as the alkyl group in the substrate alkyl halide changes from $\text{CH}_3 \rightarrow 1^\circ \rightarrow 2^\circ \rightarrow 3^\circ$?

- As the number of R groups on the carbon with the leaving group *increases*, the rate of an S_N2 reaction *decreases*.

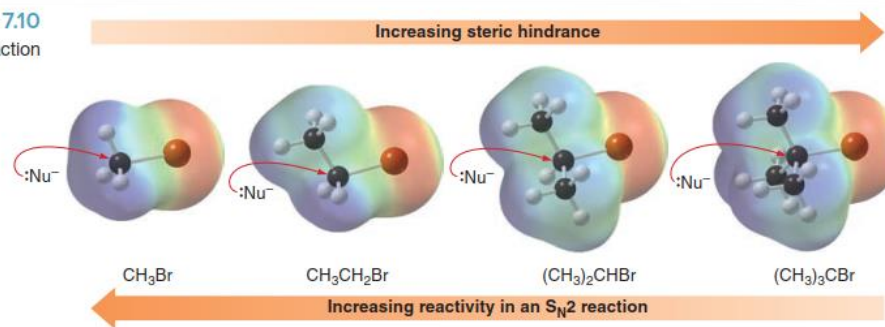


- Methyl and 1° alkyl halides undergo S_N2 reactions with ease.
- 2° Alkyl halides react more slowly.
- 3° Alkyl halides *do not* undergo S_N2 reactions.

This order of reactivity can be explained by steric effects. As small H atoms are replaced by larger alkyl groups, **steric hindrance caused by bulky R groups makes nucleophilic attack from the back side more difficult**, slowing the reaction rate. Figure 7.10 illustrates the effect of increasing steric hindrance in a series of alkyl halides.

Figure 7.10

Steric effects in the S_N2 reaction



- The S_N2 reaction is fastest with unhindered halides.

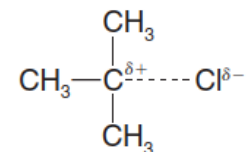
Table 7.5 summarizes what we have learned thus far about an S_N2 mechanism.

Table 7.5 Characteristics of the S_N2 Mechanism

Characteristic	Result
Kinetics	Second-order kinetics; rate = $k[\text{RX}][:\text{Nu}^-]$
Mechanism	One step
Stereochemistry	- Backside attack of the nucleophile - Inversion of configuration at a stereogenic center
Identity of R	- Unhindered halides react fastest. - Rate: $\text{CH}_3\text{X} > \text{RCH}_2\text{X} > \text{R}_2\text{CHX} > \text{R}_3\text{CX}$

6.10 A MECHANISM FOR THE S_N1 REACTION

The mechanism for the reaction of *tert*-butyl chloride with water (Section 6.9) can be described in three steps. See the box “Mechanism for the S_N1 Reaction” below, with a schematic free-energy diagram highlighted for each step. Two distinct **intermediates** are formed. The first step is the slow step—it is the rate-determining step. In it a molecule of *tert*-butyl chloride ionizes and becomes a *tert*-butyl cation and a chloride ion. In the transition state for this step the carbon–chlorine bond of *tert*-butyl chloride is largely broken and ions are beginning to develop:



The solvent (water) stabilizes these developing ions by solvation. Carbocation formation, in general, takes place slowly because it is usually a highly endothermic process and is uphill in terms of free energy.

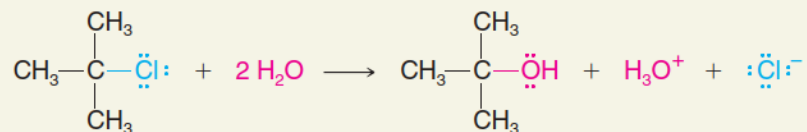
The first step requires heterolytic cleavage of the carbon–chlorine bond. Because no other bonds are formed in this step, it should be highly endothermic and it should have a high free energy of activation, as we see in the free-energy diagram. **That departure of the halide takes place at all is largely because of the ionizing ability of the solvent, water.** Experiments indicate that in the gas phase (i.e., in the absence of a solvent), the free energy of activation is about 630 kJ mol^{−1}! In aqueous solution, however, the free energy of activation is much lower—about 84 kJ mol^{−1}. **Water molecules surround and stabilize the cation and anion that are produced** (cf. Section 2.13D).

In the second step the intermediate *tert*-butyl cation reacts rapidly with water to produce a *tert*-butyloxonium ion, (CH₃)₃COH₂⁺, which in the third step, rapidly transfers a proton to a molecule of water producing *tert*-butyl alcohol.

A MECHANISM FOR THE REACTION

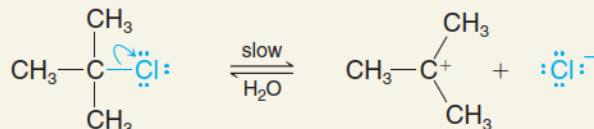
Mechanism for the S_N1 Reaction

Reaction



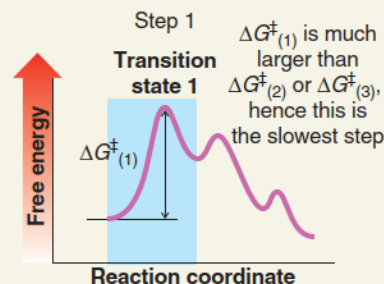
Mechanism

Step 1

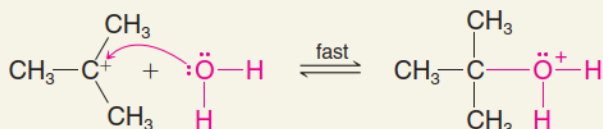


Aided by the polar solvent, a chlorine departs with the electron pair that bonded it to the carbon.

This slow step produces the 3° carbocation intermediate and a chloride ion. Although not shown here, the ions are solvated (and stabilized) by water molecules.

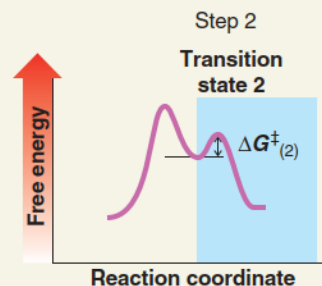


Step 2

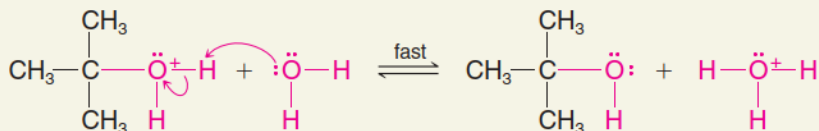


A water molecule acting as a Lewis base donates an electron pair to the carbocation (a Lewis acid). This gives the cationic carbon eight electrons.

The product is a *tert*-butyloxonium ion (or protonated *tert*-butyl alcohol).

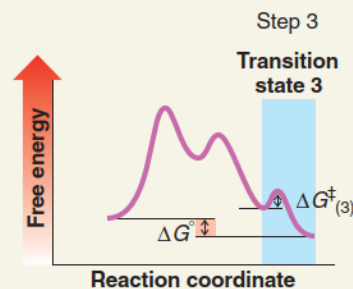


Step 3



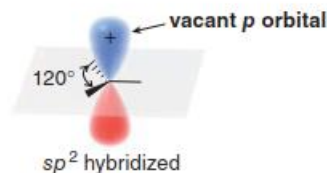
A water molecule acting as a Brønsted base accepts a proton from the *tert*-butyloxonium ion.

The products are *tert*-butyl alcohol and a hydronium ion.



7.12C Stereochemistry of the S_N1 Reaction

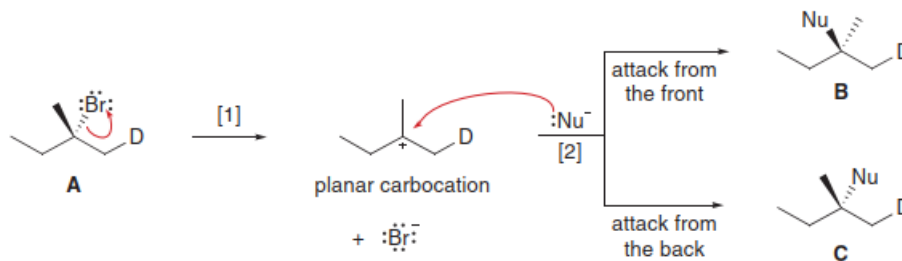
To understand the stereochemistry of the S_N1 reaction, we must examine the geometry of the carbocation intermediate.



- A carbocation (with three groups around C) is sp^2 hybridized and trigonal planar, and contains a vacant p orbital extending above and below the plane.

To illustrate the consequences of having a trigonal planar carbocation formed as a reactive intermediate, we examine the S_N1 reaction of a 3° alkyl halide **A** having the leaving group bonded to a stereogenic center.

Nucleophilic attack from both sides of a planar carbocation occurs in S_N1 reactions, but we see the result of this phenomenon only when the leaving group is bonded to a stereogenic center.



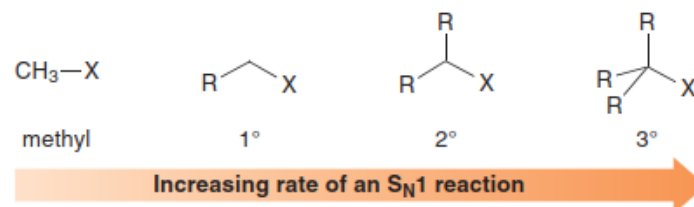
Loss of the leaving group in Step [1] generates a planar carbocation that is now achiral. Attack of the nucleophile in Step [2] can occur from either the front or the back to afford two products, **B** and **C**. These two products are *different* compounds containing one stereogenic center. **B** and **C** are stereoisomers that are not superimposable—they are **enantiomers**. Because there is no preference for nucleophilic attack from either direction, an equal amount of the two enantiomers is formed—a **racemic mixture**. We say that **racemization** has occurred.

- **Racemization** is the formation of equal amounts of two enantiomeric products from a single starting material.
- S_N1 reactions proceed with **racemization** at a single stereogenic center.

7.12D The Identity of the R Group

How does the rate of an S_N1 reaction change as the alkyl group in the substrate alkyl halide changes from $\text{CH}_3 \rightarrow 1^\circ \rightarrow 2^\circ \rightarrow 3^\circ$?

- As the number of R groups on the carbon with the leaving group *increases*, the rate of an S_N1 reaction *increases*.



- 3° Alkyl halides undergo S_N1 reactions rapidly.
- 2° Alkyl halides react more slowly.
- Methyl and 1° alkyl halides do *not* undergo S_N1 reactions.

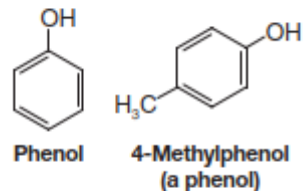
This trend is exactly opposite to that observed for the S_N2 mechanism. To explain this result, we must examine the rate-determining step, the formation of the carbocation, and learn about the effect of alkyl groups on **carbocation stability**. Table 7.6 summarizes the characteristics of the S_N1 mechanism.

Table 7.6 Characteristics of the S_N1 Mechanism

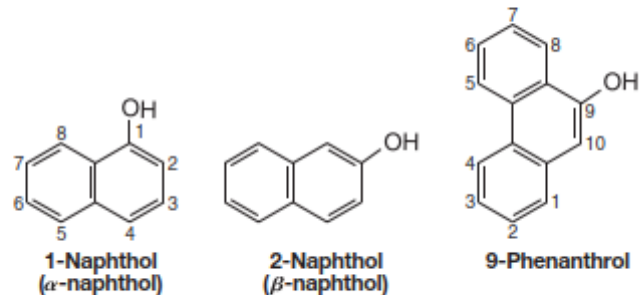
Characteristic	Result
Kinetics	First-order kinetics; rate = $k[\text{RX}]$
Mechanism	Two steps
Stereochemistry	- Trigonal planar carbocation intermediate - Racemization at a single stereogenic center
Identity of R	- More substituted halides react fastest. - Rate: $\text{R}_3\text{CX} > \text{R}_2\text{CHX} > \text{RCH}_2\text{X} > \text{CH}_3\text{X}$

21.1 STRUCTURE AND NOMENCLATURE OF PHENOLS

Compounds that have a hydroxyl group directly attached to a benzene ring are called **phenols**. Thus, **phenol** is the specific name for hydroxybenzene, and it is the general name for the family of compounds derived from hydroxybenzene:

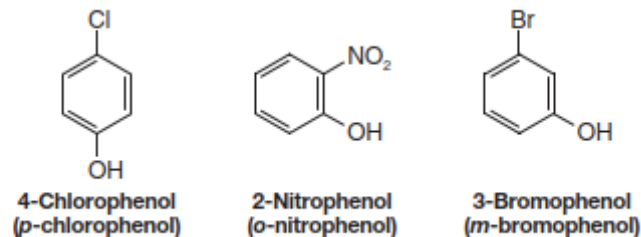


Compounds that have a hydroxyl group attached to a polycyclic benzenoid ring are chemically similar to phenols, but they are called **naphthols** and **phenanthrols**. For example:

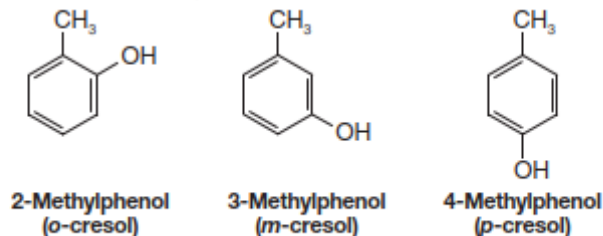


21.1A Nomenclature of Phenols

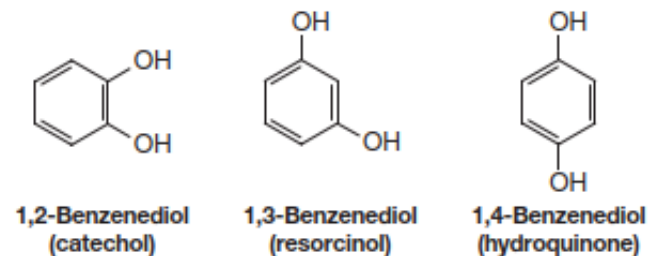
We studied the nomenclature of some phenols in Chapter 14. In many compounds *phenol* is the base name:



The methylphenols are commonly called *cresols*:



The benzenediols also have common names:



21.3 PHYSICAL PROPERTIES OF PHENOLS

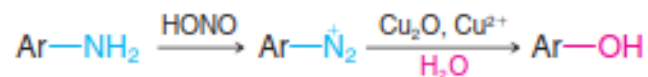
The presence of hydroxyl groups in phenols means that phenols are like alcohols (Section 11.2) in some respects. For example, they are able to form strong intermolecular hydrogen bonds, and therefore have higher boiling points than hydrocarbons of the same molecular weight. Phenol (bp 182 °C) has a boiling point more than 70 °C higher than toluene (bp 110.6 °C), even though the two compounds have almost the same molecular weight. Phenols are also modestly soluble in water because of their ability to form strong hydrogen bonds with water molecules.

21.4 SYNTHESIS OF PHENOLS

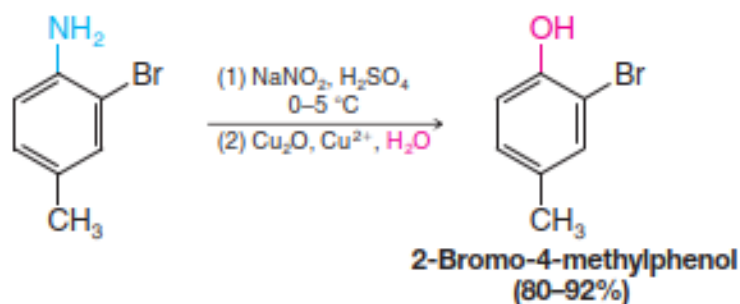
21.4A Laboratory Synthesis

The most important laboratory synthesis of phenols is by hydrolysis of arenediazonium salts (Section 20.7E). This method is highly versatile, and the conditions required for the diazotization step and the hydrolysis step are mild. This means that other groups present on the ring are unlikely to be affected.

General Reaction



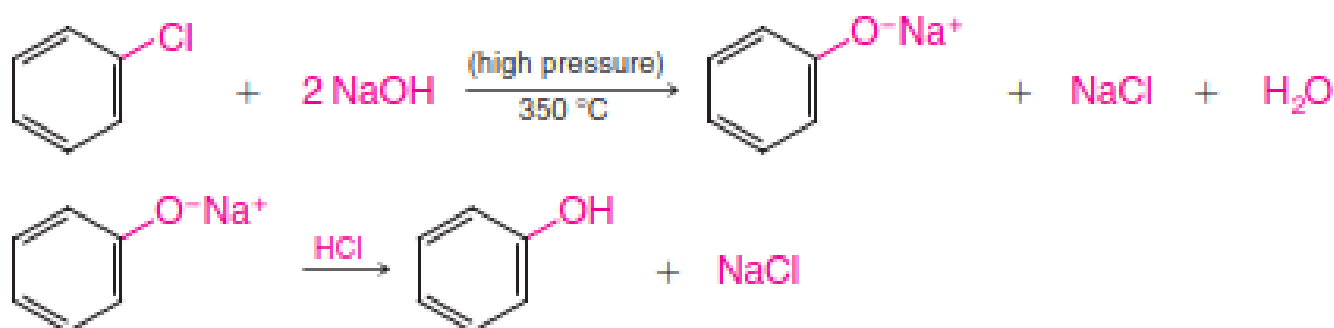
Specific Example



21.4B Industrial Syntheses

Phenol is a highly important industrial chemical; it serves as the raw material for a large number of commercial products ranging from aspirin to a variety of plastics. Worldwide production of phenol (which in industry is sometimes called carbolic acid) is more than 3 million tons per year. Several methods have been used to synthesize phenol commercially.

- 1. Hydrolysis of Chlorobenzene (Dow Process).** In this process chlorobenzene is heated at 350 °C (under high pressure) with aqueous sodium hydroxide. The reaction produces sodium phenoxide, which, on acidification, yields phenol. The mechanism for the reaction probably involves the formation of benzyne (Section 21.11B).



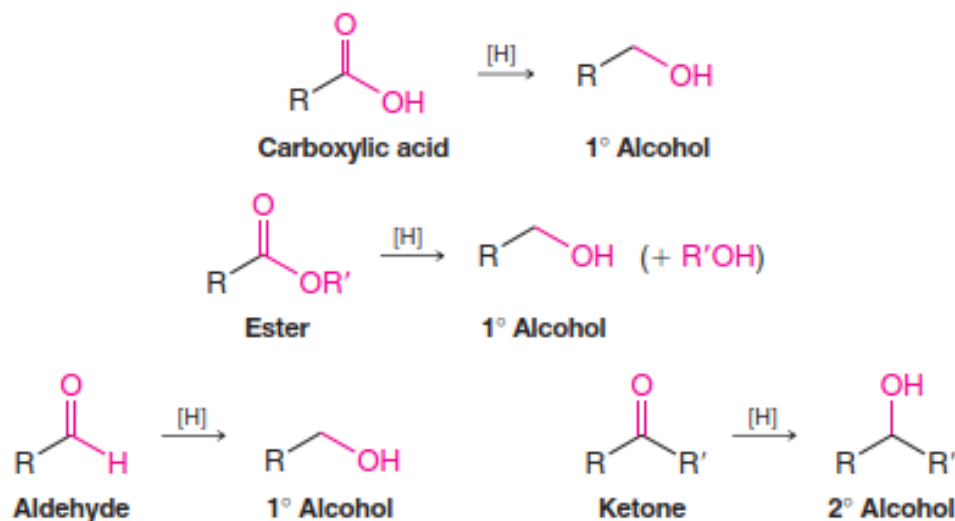
12.3 ALCOHOLS BY REDUCTION OF CARBONYL COMPOUNDS

Photo courtesy of Aldrich
Chemical Co.



Unless special precautions are taken, lithium aluminum hydride reductions can be very dangerous. You should consult an appropriate laboratory manual before attempting such a reduction, and the reaction should be carried out on a small scale.

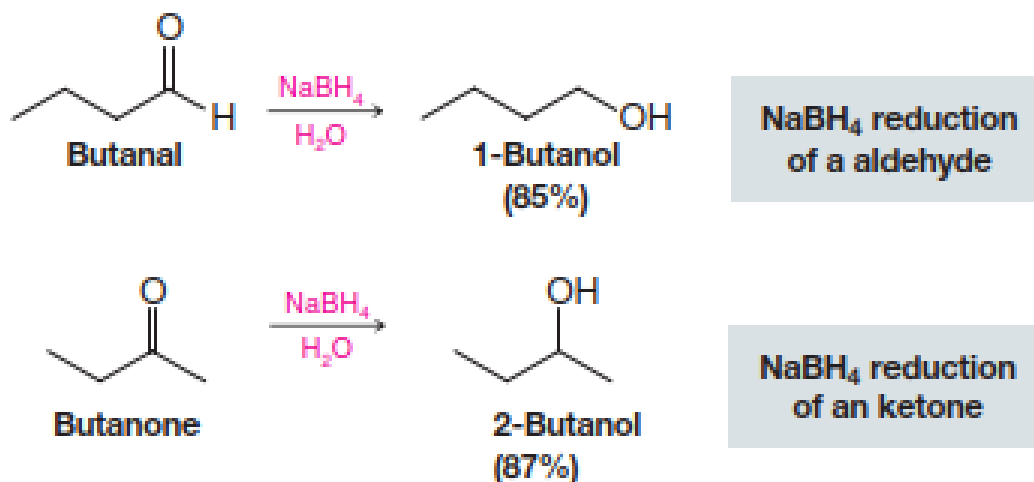
Primary and secondary alcohols can be synthesized by the **reduction** of a variety of compounds that contain the carbonyl group. Several general examples are shown here:



12.3B Sodium Borohydride

- Aldehydes and ketones are easily reduced by sodium borohydride (NaBH_4).

Sodium borohydride is usually preferred over LAH for the reduction of aldehydes and ketones. Sodium borohydride can be used safely and effectively in water as well as alcohol solvents, whereas special precautions are required when using LAH.



Aldehydes and ketones can be reduced using hydrogen and a metal catalyst, as well, and by sodium metal in an alcohol solvent.

The key step in the reduction of a carbonyl compound by either lithium aluminum hydride or sodium borohydride is the transfer of a **hydride ion** from the metal to the carbonyl carbon. In this transfer the hydride ion acts as a *nucleophile*. The mechanism for the reduction of a ketone by sodium borohydride is illustrated here.